

University of Colombo School of Computing
SCS 3312 : Research Methods

D4: Final Research Report

*“An Investigation into Student Behavior During Free Periods Between
Scheduled Lectures at UCSC”*

Student Name and Index

N.P.J. Chathusara 23000244

Date of Submission

03 August 2026

Acknowledgement

I would like to express my sincere gratitude to the lecturer of SCS3312 - Research Methods, Dr. Manjusri Wickramasinghe, for his continued guidance and support throughout every phase of this research project.

I would also like to thank my teammates, A.S. Haputhantri and W.M.V.A.N. Amasha, for their contributions to the fieldwork, survey design, and data collection that this final report draws upon. I am also grateful to the students who volunteered for the earlier observations and interviews during the fieldwork phase, and to the UCSC undergraduate students who gave their time to complete the survey instrument at the five field sites; their honest participation made both the qualitative and quantitative phases of this study possible.

Abstract

University students have regular gaps between their scheduled lectures, but little is known about how they actually use this time, or whether the campus spaces available to them meet their needs. Understanding this can help the university plan better spaces and support student wellbeing. This report is the final, integrated submission for a study conducted in three phases at the University of Colombo School of Computing (UCSC): structured observations and interviews at five campus locations — Library, Canteen, Dewale, Common Room, and Open Area (D1); the design and piloting of a survey instrument (D2); and a full-scale survey of 60 students across the same five locations (D3). This report combines the qualitative and quantitative evidence from all three phases into a single analysis of student behavior during free periods.

The findings show that students choose locations based on the activity they plan to do rather than habit, and that free-period length is the strongest single driver of both activity and location choice. Certain campus spaces are avoided mainly for environmental reasons such as smell, heat, and crowding, while some student groups face a genuine structural access gap to particular spaces. Seating availability was the most consistently poorly rated aspect of campus facilities.

These findings, drawn from combining direct fieldwork with a 60-student survey, provide a more complete and reliable account of student behavior than either method alone. They also point to specific, actionable priorities for campus space planning, and reflect on the strengths and limitations of the mixed-methods design used throughout the study.

Contents

Acknowledgement	i
Abstract	ii
Table of Contents	iii
List of Tables	iv
List of Acronyms	v
1 Introduction	1
2 Research Approach: From Fieldwork to Survey	2
3 Key Findings	3
3.1 Location choice follows activity, not habit	3
3.2 Free-Period Duration is the central driver of the behavior	3
3.3 Avoidance by choice vs. Avoidance by Restriction	4
3.4 Phone use is common but situational	4
3.5 Satisfaction is Driven by Facilities, Not Access	4
4 Reconciling the Qualitative and Quantitative Evidence	5
5 Implications for Real-World Campus Space Planning	6
6 Critical Analysis	7
6.1 Reflection on research design	7
6.2 Strengths and limitations	7
7 Conclusion and Future Work	9
7.1 Conclusion	9
7.2 Future Work	9
Index	11

List of Tables

6.1	Key strengths and limitations of the research approach	8
-----	--	---

List of Acronyms

UCSC University of Colombo School of Computing

1 Introduction

This report is the final individual submission for the research, An Investigation into Student Behavior During Free Periods Between Scheduled Lectures at UCSC. In this study, a free period is defined as an unscheduled period between two timetabled lectures or practicals in the academic day. Time made available because of students choosing not to attend is not considered a free period. At UCSC, the sessions are scheduled from 8.00 am to 12.00 pm in the morning and 1.00 pm to 5.00 pm in the afternoon. Therefore, the study excludes the lunch break (12.00 pm–1.00 pm) and the time before a student’s first session and after their last session of the day. The study was conducted in five typical places where students spend this period on campus. These places include Library, Canteen, Dewale, Common Room and Open Area.

It brings together the results of the three research stages: D1, through structured observations and short interviews conducted in these five campus locations; D2, by designing and piloting the survey questionnaire; and D3, by collecting answers from 60 undergraduate students through a quantitative survey.

The primary goal of the study was to determine how UCSC students spend their free time between planned lectures and whether the current campus areas meet their needs. This paper summarizes the general findings from both qualitative and quantitative phases, contrasts the survey findings with the earlier field observations, and examines what the combined findings suggest for improving the layout and usage of campus areas.

2 Research Approach: From Fieldwork to Survey

The study employed a two-phase mixed methods design, with each step informing the subsequent phase. In D1, structured observations and semi-structured interviews were conducted at five campus locations to understand how students spend their free periods. The findings from this qualitative phase were used to develop the D2 survey.

The findings led D2 to develop a 19-question survey that asked participants to provide information on five topics: participant background, space use and avoidance, factors influencing location choice, activities during free periods, and satisfaction with campus facilities. The survey was piloted on 10 students and the problems of phrasing and formatting were resolved before the final version was made. The survey was completed by 60 students in D3 (14 first-years, 21 second-years, 21 third-years and 4 fourth-years) at the same five sites. Descriptive statistics and basic cross-tabulations were used to analyze answers. As each survey question was built from the D1 findings, it was possible to compare the quantitative results directly with the previous field observations and not consider it as a separate dataset.

3 Key Findings

3.1 Location choice follows activity, not habit

The library was the most popular campus space, with 46.7% of respondents selecting it as their primary location and 86.7% including it among their top three. This supports the D1 finding that students choose locations based on the activity they intend to perform. Likewise, the statement "I prefer locations that match the activity I plan to do" received the second-highest agreement score (mean = 4.07), indicating the importance of activity suitability in location selection.

The second most common principal places were Dewale (23.3%) and Open Area (20.0%). The Common Room was chosen less often, indicating that it is used by a smaller group of regular users. None of the respondents chose the Canteen as their main location but it was in the top three of more than half of the respondents. This shows that the Canteen is not a main destination but is used for dining and socializing. Some of the fourth-year respondents also included "meeting with project supervisors" as an activity not covered in the survey, implying that their leisure periods are spent more on research work and in research laboratories rather than the five main campus areas. This further reinforces the fact that students choose sites based on the activities they need to do.

3.2 Free-Period Duration is the central driver of the behavior

The length of a free period had a considerable influence on the student's activities and choice of location. Students having free periods of less than 30 minutes put more significance on being close to their next lecture hall (mean = 4.00) although this value dropped for free periods longer than two hours (mean = 2.24). Shorter free periods were mainly spent on eating, talking with friends, or browsing phones, while longer periods were more often used for studying or amusement.

The statement measuring the influence of free-period length on location choice showed the greatest variation among respondents (mean = 3.61, SD = 1.40), indicating that its influence differs across individuals. Collectively, the results support the

D1 finding that the length of a free period affects activity and location choices.

3.3 Avoidance by choice vs. Avoidance by Restriction

More than half of the respondents (56.7%) avoided at least one campus location. The Canteen was mainly avoided because of unpleasant smells and overcrowding, while the Open Area was avoided due to heat and crowding, closely matching the D1 findings.

A few respondents also mentioned bad odours in the Dewale, an aspect which was not found in D1 and is thus worth additional examination. The study also made a distinction between voluntary avoidance and restricted access. All first-year responders said they were restricted to use Dewale and had no Common Room reserved for them. This is a limitation of campus provision, not a personal preference, hence it is crucial to study this apart from usual avoidance.

3.4 Phone use is common but situational

For 31.7% of respondents phone browsing was in the top three activities but it was the main activity for just 6.7% which is consistent with D1 finding that it is frequently a secondary activity and not a primary one. When asked whether phone browsing was a worthwhile use of free time, 85% of respondents selected "neutral" or "depends on the situation," indicating that most students did not hold a strong opinion about it.

The relationship between spending time with friends and phone browsing was weak ($r = 0.16$), suggesting that the survey could not fully capture the differences between individual and group phone use that were observed during D1.

3.5 Satisfaction is Driven by Facilities, Not Access

The most major issue with facilities was the availability of sitting, which was rated lowest in the survey (mean = 2.46). Over half of all respondents (55%) stated that they did not have sufficient space during peak hours. Support for varied activities (mean = 2.85) was also below average, showing that available venues did not always match students' needs.

4 Reconciling the Qualitative and Quantitative Evidence

Comparing the D1 fieldwork against the D3 survey results show a clear pattern of confirmation, with some findings holding up more precisely than others:

Strongly Confirmed at scale:

- **Location activity matching** — Students choose spaces based on the task they plan to do, not habit.
- **Free period duration** — The length of a free period shapes both where students go and what they do.
- **Environmental filtering** — Heat, noise, and crowding influence space choice (this item scored highest of all Section C statements from the survey, mean 4.10/5).

Only partially confirmed:

- **Context-dependent phone use** — the survey confirmed the existence of the pattern but not to the same degree of precision as direct observation.
- **Solo vs. group space occupancy** — similarly present, but not measurable at the level of detail D1's fieldwork achieved. For example, D1 observed directly that solo students engaged more with phones and left sooner than groups, but the survey only measured frequency of each behavior separately, not how they interact together in the same moment.

5 Implications for Real-World Campus Space Planning

The findings point to specific, addressable priorities rather than a general shortage of space.

- **Increase seating capacity during peak hours:** The study found the main problem as the availability of seating, contributing to dissatisfaction across all year groups. Increasing the number of available seats during busy periods is likely to have the greatest positive impact on students' experiences.
- **Improve canteen facilities:** The main reasons students avoided the Canteen were unpleasant smells and overcrowding. Improving ventilation, maintaining better hygiene, and managing crowding could make the room more comfortable without compromising the basic purpose of the space.
- **Focus on improving the quality of facilities, rather than simply increasing access:** The results indicated that the third-year students, who had access to all the main spaces on campus, were the least satisfied. Thus, improving the quality and applicability of current facilities is likely to be more useful than merely providing new places. For instance, the low score for activity support (E3) also shows spaces should better support the diversity of activities students are presently engaged in.
- **Consider providing a dedicated space for unwell students:** This need was not reflected in the survey data as no survey question expressly asked about it but was highlighted in the D1 interviews. Although UCSC has a medical center that provides consultations and treatment, there is no place nearby where students who feel briefly ill can go and lay down or rest, between classes without needing medical assistance.

6 Critical Analysis

6.1 Reflection on research design

An important analytical choice in this study was to differentiate structural access constraints from preference-based avoidance. This allowed identifying the access gap for 1st-year students other than voluntary avoidance to places like Canteen or Open Area owing to environmental circumstances. Without this distinction, these fundamentally different issues would have been treated in the same way, making it difficult to identify the need for different solutions. More generally, this is an example of the study's position that one source of data, whether an interview or a survey response, is typically insufficient to judge whether a behavior is due to personal preference, circumstances, or access constraints.

The two-phase study design also incorporated the strengths of both qualitative and quantitative methodologies. Observations and interviews with D1 gave rich details about student behavior that could not be captured by a closed ended survey question. For example, students talked about the context in which they used their phones and the need for space to rest when they were unwell. D3 next looked to see whether these trends were present in a wider sample of students. Neither method alone would have been sufficient: fieldwork allows us to gain deep insights from a small number of participants, whereas surveys allow us to measure patterns from a wider sample, but only within the boundaries of the questions asked.

6.2 Strengths and limitations

Strengths	Limitations
Survey questions were derived directly from the D1 observations, enabling a comparison between the quantitative and qualitative results.	The fourth-year sample was small ($n = 4$), so findings for this group should be interpreted with caution.

Strengths	Limitations
Facility access issues were identified more easily by analyzing structural access restrictions separately from voluntary avoidance.	The survey did not differentiate between year specific Common Rooms; an assumption was therefore necessary for the study.
Utilizing both qualitative and quantitative methodologies provides a more thorough knowledge of student behavior during free periods.	Data was collected during a single non-exam period, so student behavior may differ during examinations or assignment periods.
The mixed-methods approach allowed the survey findings to validate, refine, and extend the D1 field observations.	The survey quantified the frequency of behaviours but was unable to capture the context or purpose, and some replies may have been subject to self-report bias.

Table 6.1: Key strengths and limitations of the research approach

7 Conclusion and Future Work

7.1 Conclusion

Across both phases of this research, UCSC students show a clear and largely deliberate logic in how they spend their free periods: they match location to task, let the length of their free time guide both activity and location choice. They are also influenced by the quality and comfort of campus spaces.

The investigation also revealed some shortcomings in the current facilities. The D3 study validated many of the findings from the D1 survey but it also found an access gap for first year students and a very low satisfaction level amongst third year students despite their complete access to campus area. Overall the results show that the free-period experience is enhanced not just by greater access to campus locations but through better seating, excellent facilities and practical student needs.

7.2 Future Work

Several results in this study point out specific questions worth investigating further rather than being treated as settled:

- Conduct follow-up observations to investigate the reports of unpleasant smells in the Dewale, as this issue was not identified during the D1 fieldwork.
- Carry out more qualitative study to better understand the relationship between phone use and social setting that was not fully reflected in the survey.
- Investigate the factors contributing to the unexpectedly low satisfaction reported by third-year students despite their unrestricted access to campus spaces.
- Using year-qualified Common Room questions in future surveys would help resolve the ambiguity noted in this report's limitations.
- Conduct a follow-up study of the Dewale to assess how its recent renovation, including the addition of food services and increased staff presence, has affected

student behavior, space usage, and the location's previously observed role as a quiet academic space.

Index

Canteen, 1, 3, 4

Common Room, 1, 3

Dewale, 1, 4

free period, 1, 3

Library, 1

Open Area, 1, 4